

A large, ancient tree with a thick trunk and a wide, spreading canopy of green leaves stands in a lush green valley. In the background, rolling hills and mountains are visible under a blue sky with scattered white clouds. The scene is peaceful and scenic.

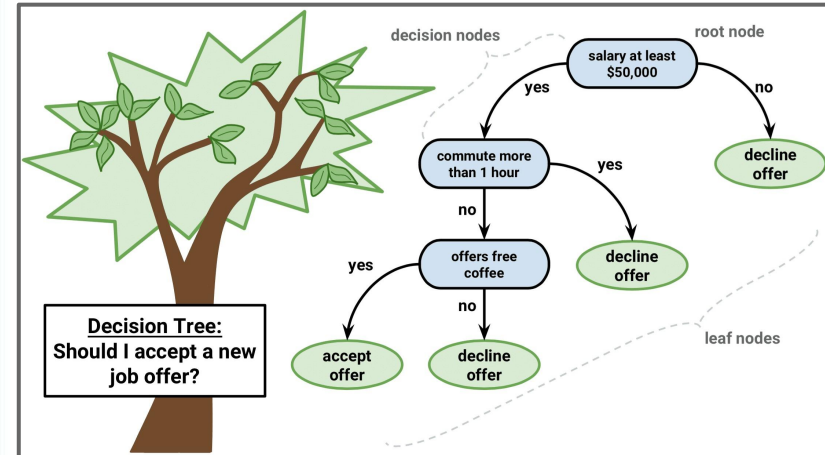
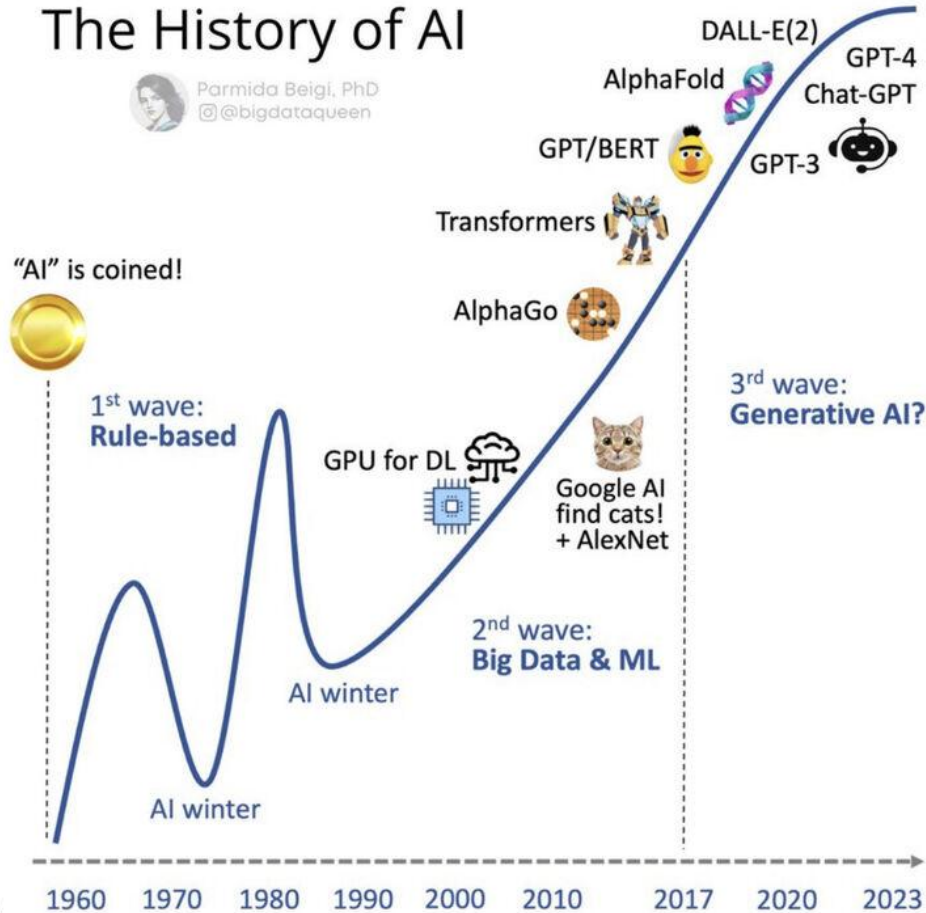
Decision Tree

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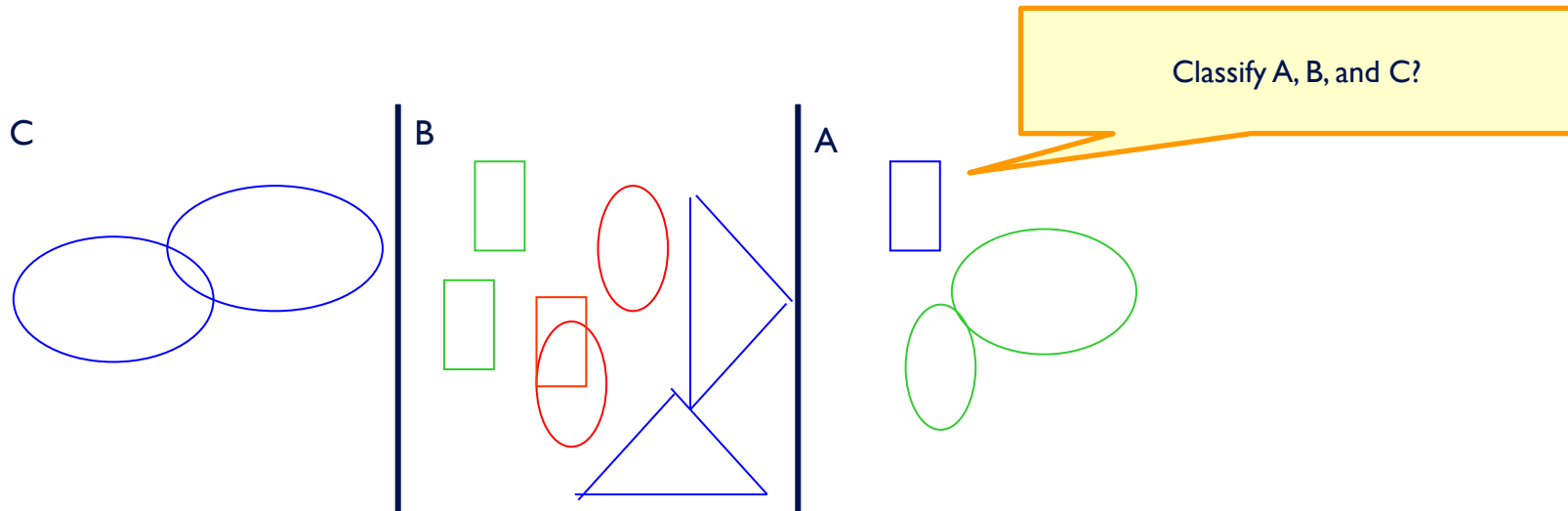
The History of AI



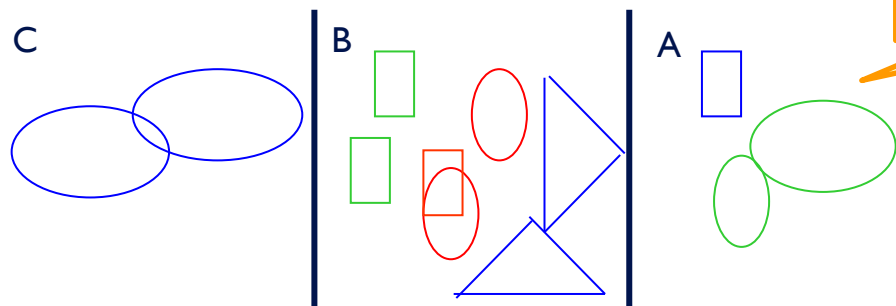
Parmida Beigi, PhD
@bigdataqueen



Motivation

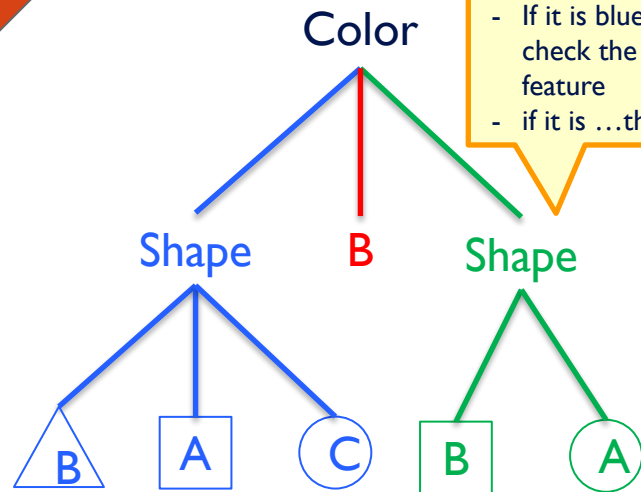
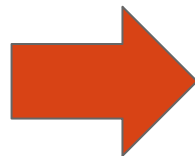


Motivation



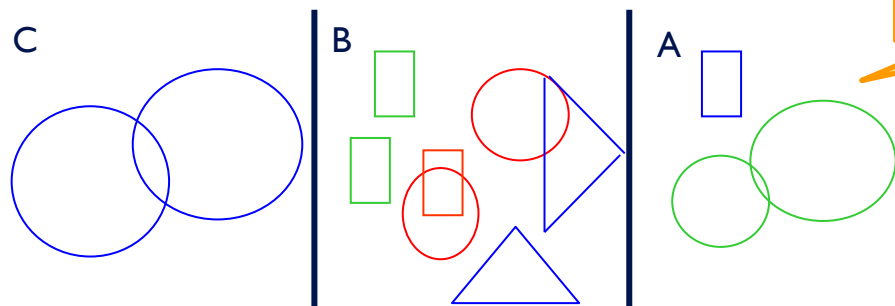
Given this collection of shapes, what shapes are type A, B, and C?

Learning a Decision Tree?



- Check the color feature.
- If it is blue then check the shape feature
- if it is ...then...

Motivation

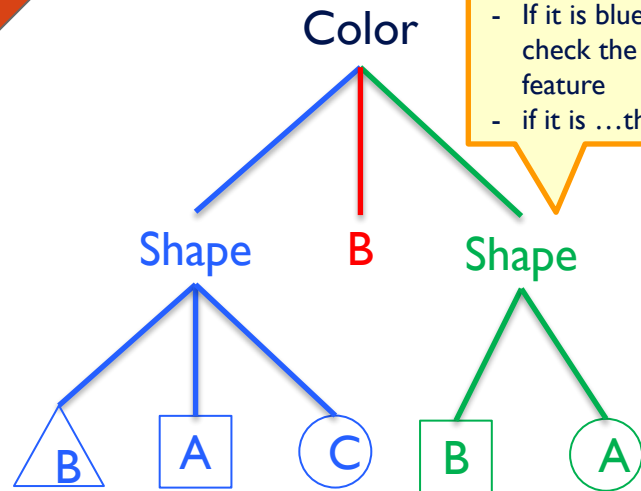


Given this collection of shapes, what shapes are type A, B, and C?

Learning a Decision Tree?

- Check the color feature.
- If it is blue then check the shape feature
- if it is ...then...

- Red = B
- Green $\wedge$ Square = B
- Green $\wedge$ Circle = A
- Blue $\wedge$ Triangle = B
- Blue $\wedge$ Square = A
- Blue $\wedge$ Circle = C



Decision Tree Elements

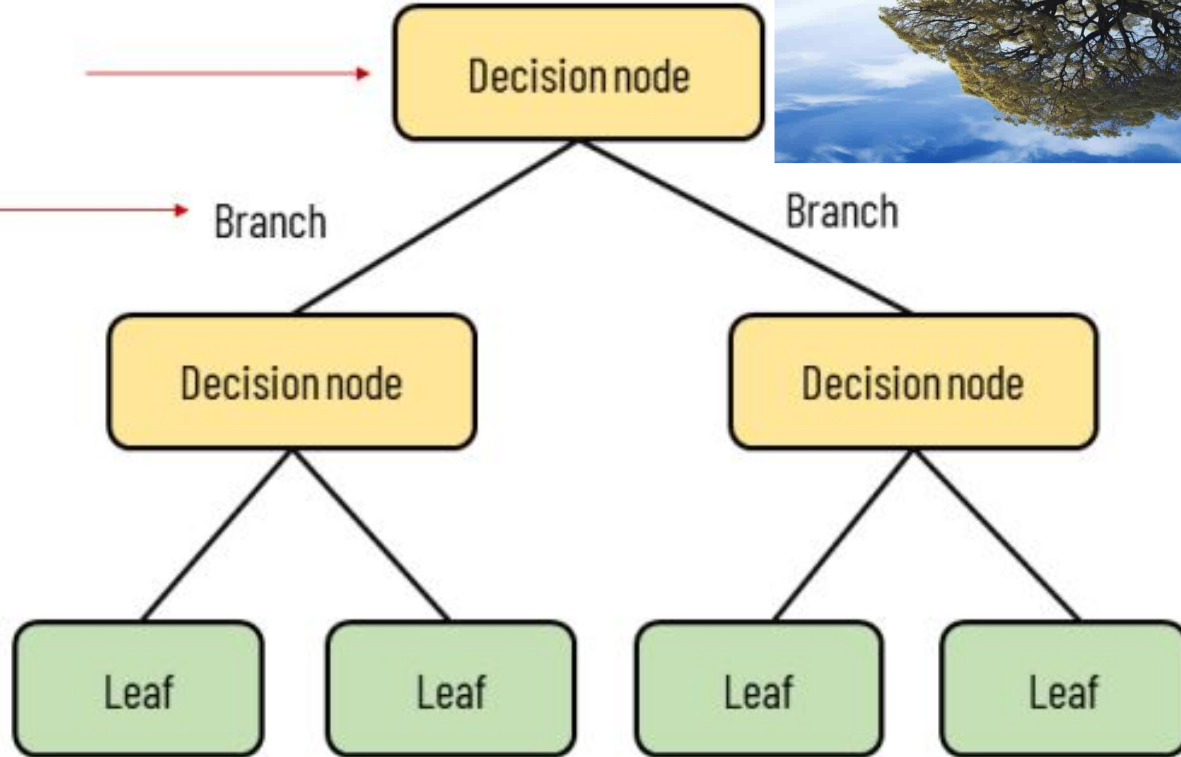
Root
node



Condition (choice)

Alternatives

Decisions
(outcomes)



Sample Dataset

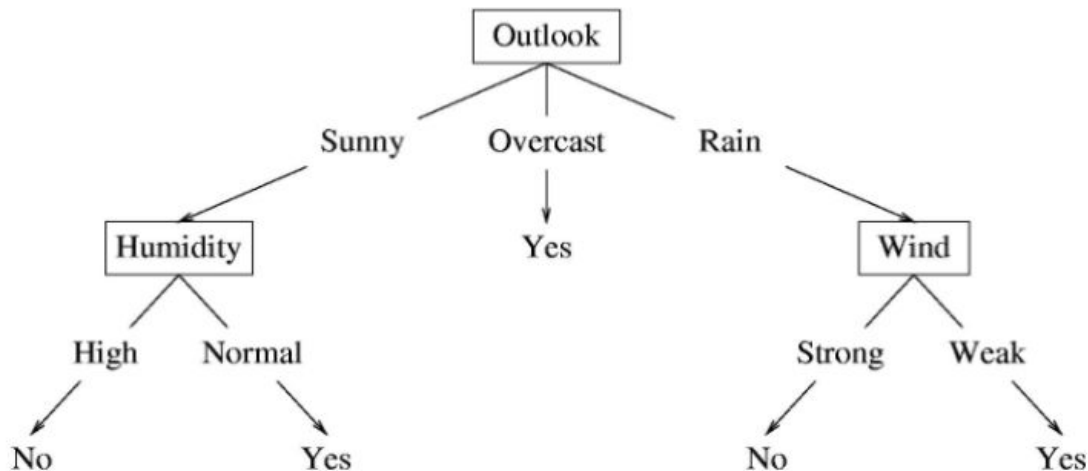
- Columns denote features X_i
- Rows denote labeled instances $\langle x_i, y_i \rangle$
- Class label denotes whether a tennis game was played

$\langle x_i, y_i \rangle$

Predictors				Response
Outlook	Temperature	Humidity	Wind	Class
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Decision Tree

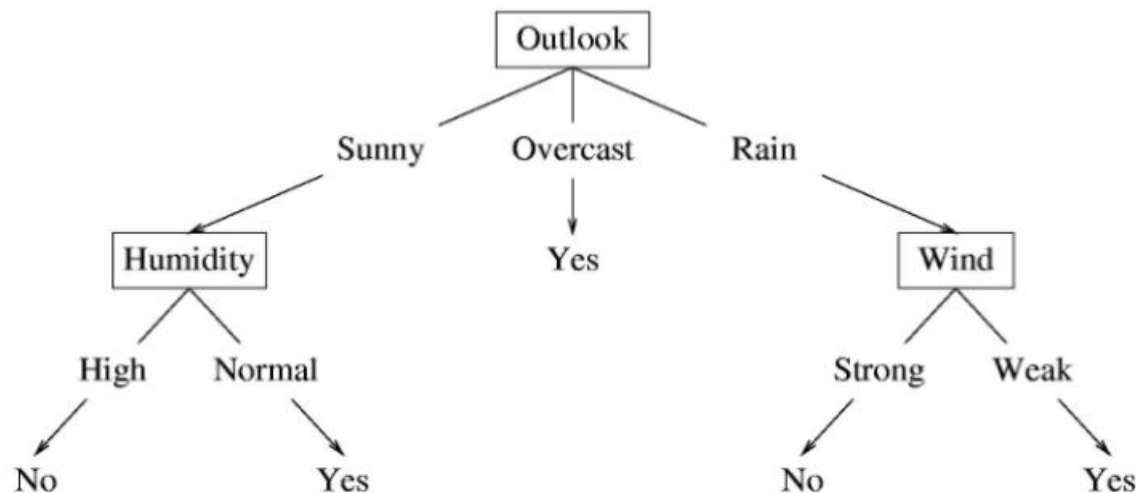
- A possible decision tree for the data:



- Each internal node: test one attribute X_i
- Each branch from a node: selects one value for X_i
- Each leaf node: predict Y (or $p(Y \mid \mathbf{x} \in \text{leaf})$)

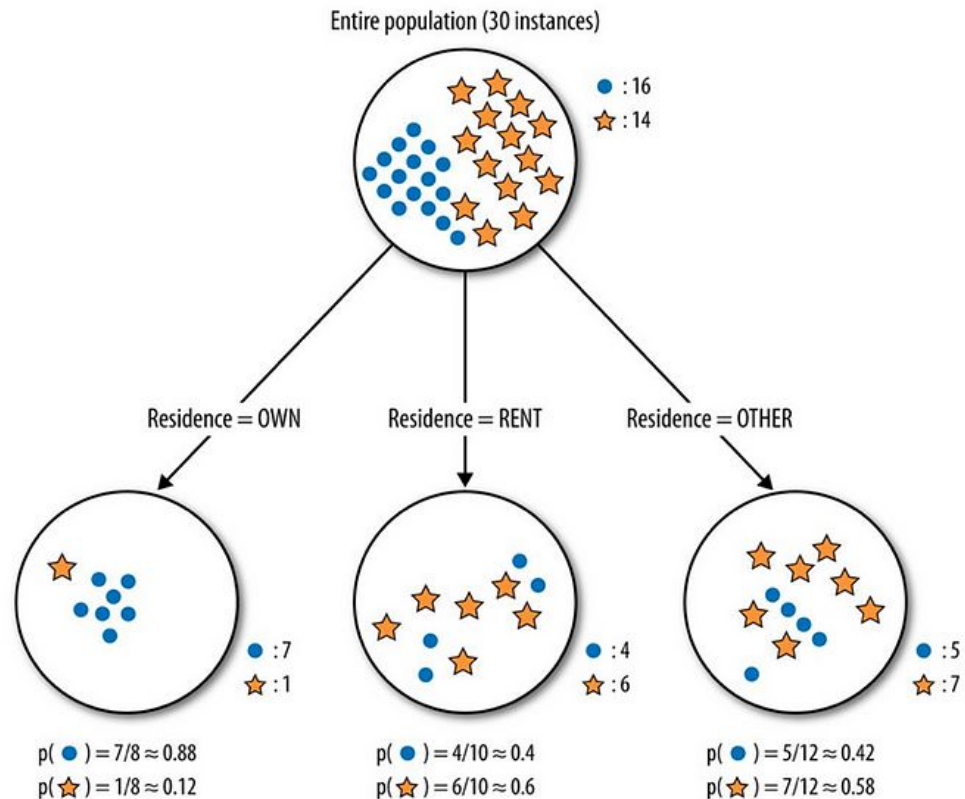
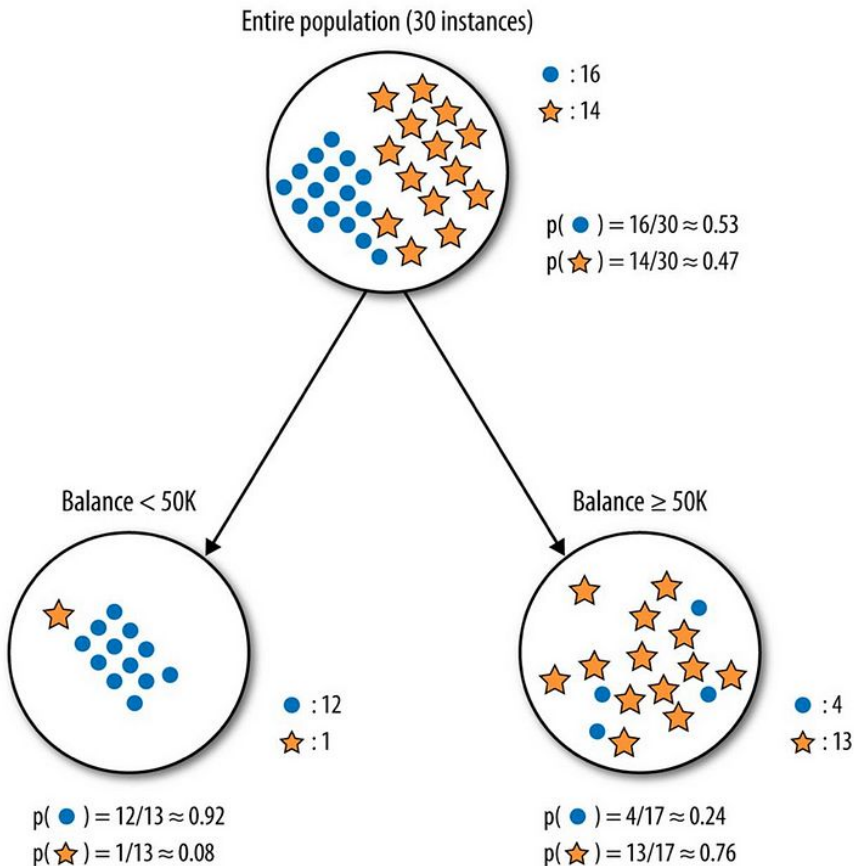
Decision Tree

- A possible decision tree for the data:



- What prediction would we make for
<outlook=sunny, temperature=hot, humidity=high, wind=weak> ?

Decision Tree Modeling



Impurities: When to split a Node?

Very Impure



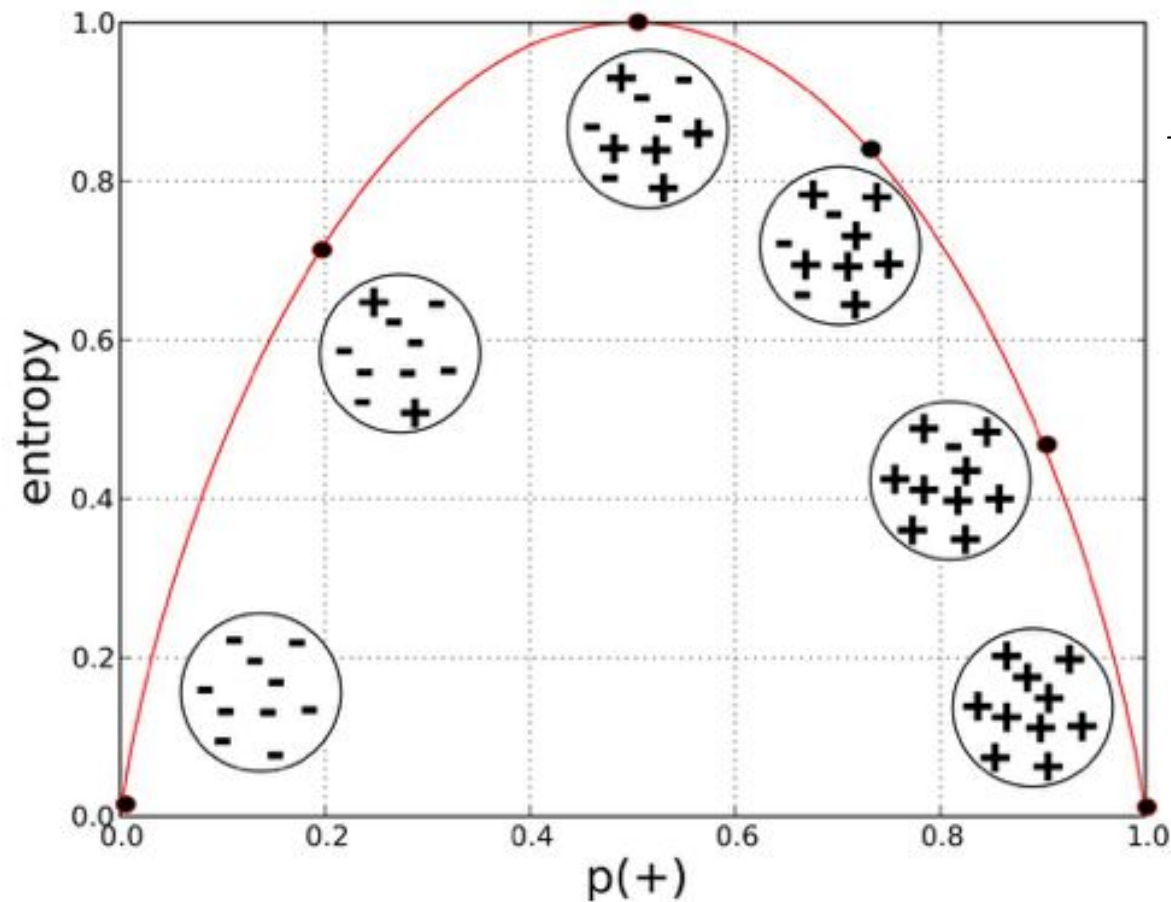
Less Impure



Pure

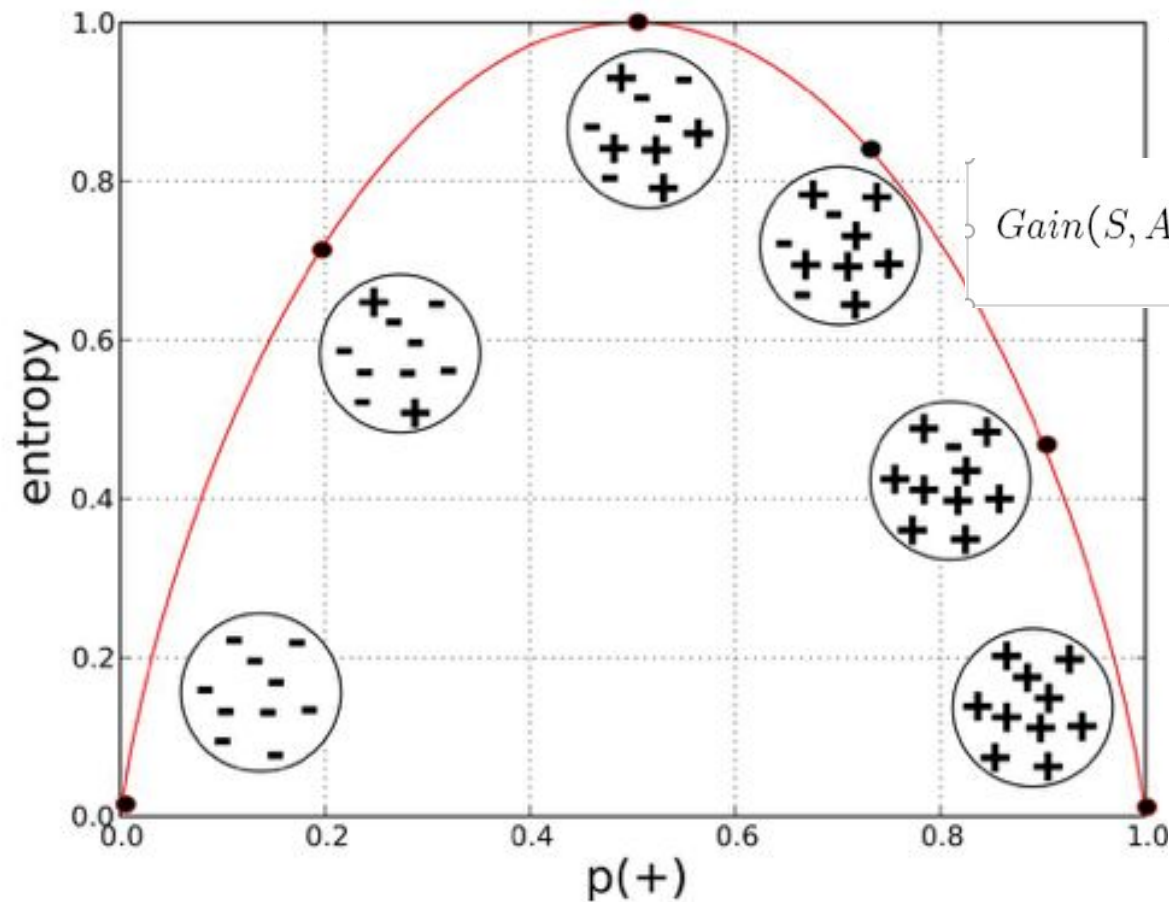


Impurity Measure: Entropy



$$H = - \sum_{i=1}^N p_i \log(p_i)$$

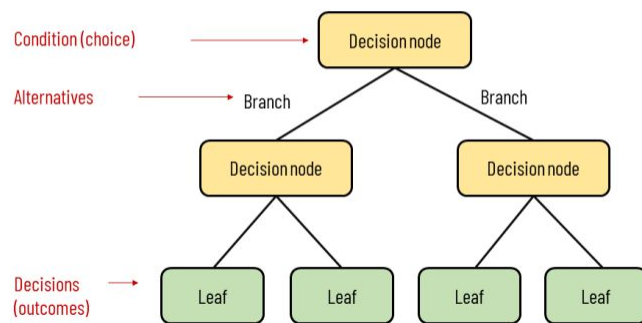
Impurity Measure: Entropy



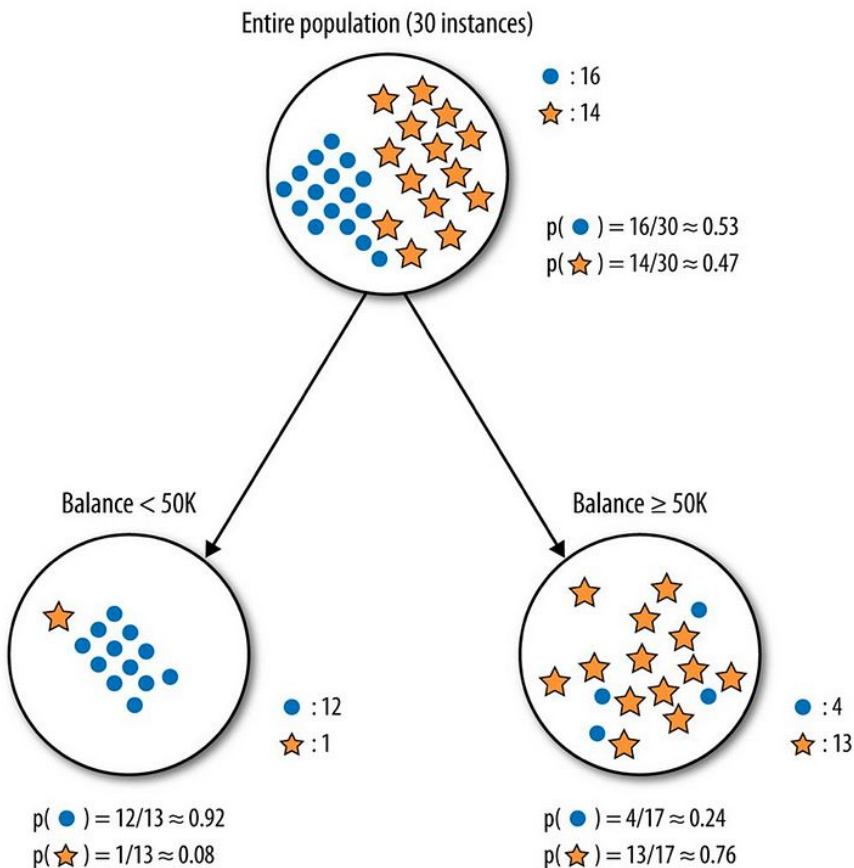
$$H = - \sum_{i=1}^N p_i \log(p_i)$$

$$Gain(S, A) \equiv Entropy(S) - \sum_{v \in D_A} \frac{|S_v|}{|S|} Entropy(S_v)$$

Elements of a decision tree



Decision Tree Modeling



$$E(\text{Parent}) = -\frac{16}{30} \log_2 \left(\frac{16}{30} \right) - \frac{14}{30} \log_2 \left(\frac{14}{30} \right) \approx 0.99$$

$$E(\text{Balance} < 50K) = -\frac{12}{13} \log_2 \left(\frac{12}{13} \right) - \frac{1}{13} \log_2 \left(\frac{1}{13} \right) \approx 0.39$$

$$E(\text{Balance} \geq 50K) = -\frac{4}{17} \log_2 \left(\frac{4}{17} \right) - \frac{13}{17} \log_2 \left(\frac{13}{17} \right) \approx 0.79$$

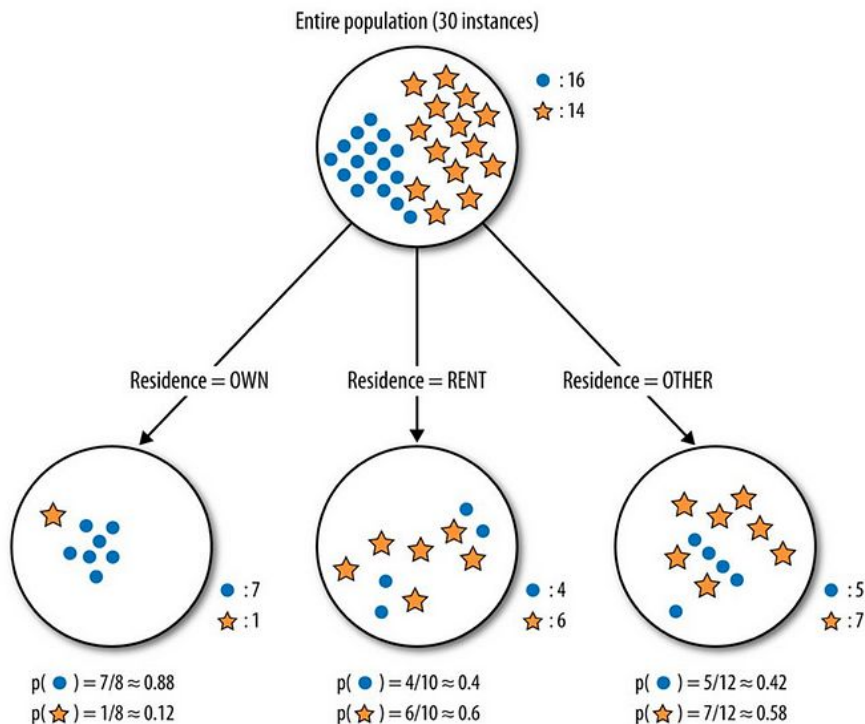
Weighted Average of entropy for each node:

$$\begin{aligned} E(\text{Balance}) &= \frac{13}{30} \times 0.39 + \frac{17}{30} \times 0.79 \\ &= 0.62 \end{aligned}$$

Information Gain:

$$\begin{aligned} IG(\text{Parent}, \text{Balance}) &= E(\text{Parent}) - E(\text{Balance}) \\ &= 0.99 - 0.62 \\ &= 0.37 \end{aligned}$$

Decision Tree Modeling



$$E(\text{Residence} = \text{OWN}) = -\frac{7}{8}\log_2\left(\frac{7}{8}\right) - \frac{1}{8}\log_2\left(\frac{1}{8}\right) \approx 0.54$$

$$E(\text{Residence} = \text{RENT}) = -\frac{4}{10}\log_2\left(\frac{4}{10}\right) - \frac{6}{10}\log_2\left(\frac{6}{10}\right) \approx 0.97$$

$$E(\text{Residence} = \text{OTHER}) = -\frac{5}{12}\log_2\left(\frac{5}{12}\right) - \frac{7}{12}\log_2\left(\frac{7}{12}\right) \approx 0.98$$

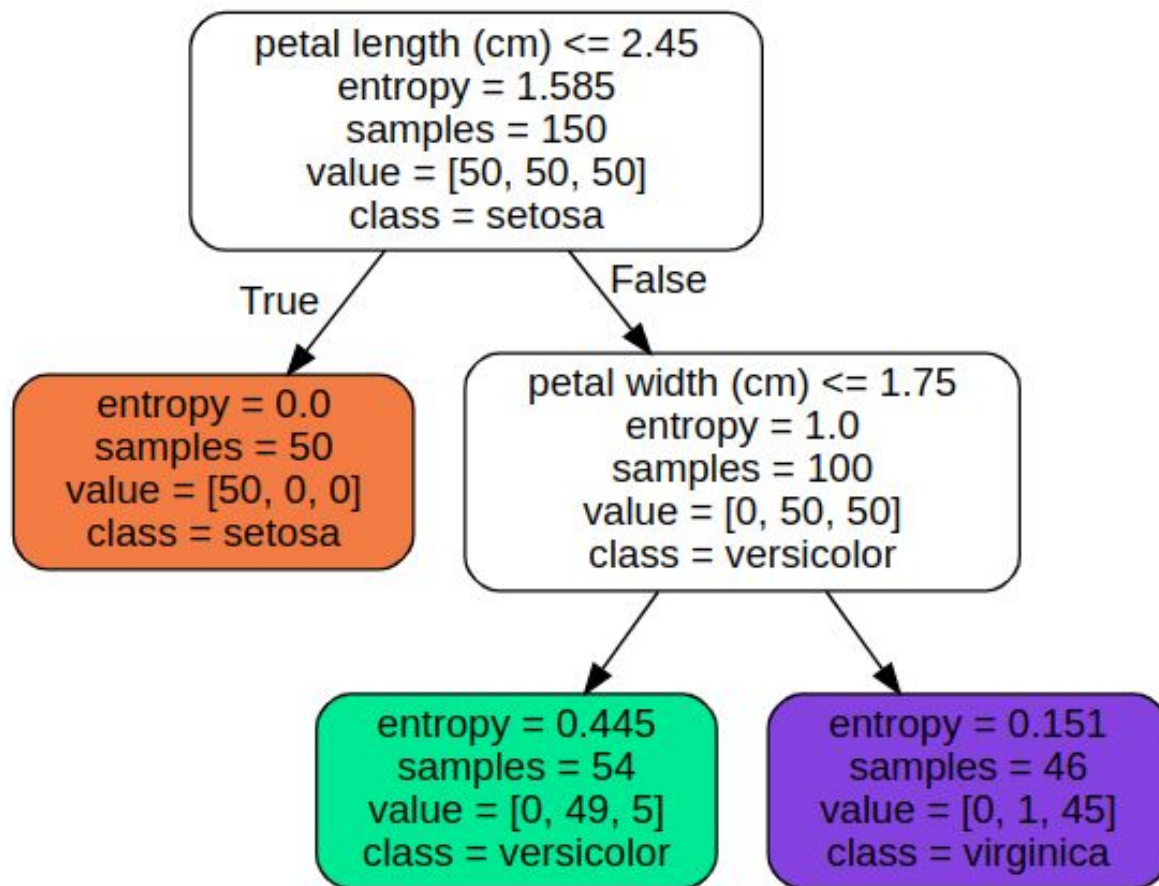
Weighted Average of entropies for each node:

$$E(\text{Residence}) = \frac{8}{30} \times 0.54 + \frac{10}{30} \times 0.97 + \frac{12}{30} \times 0.98 = 0.86$$

Information Gain:

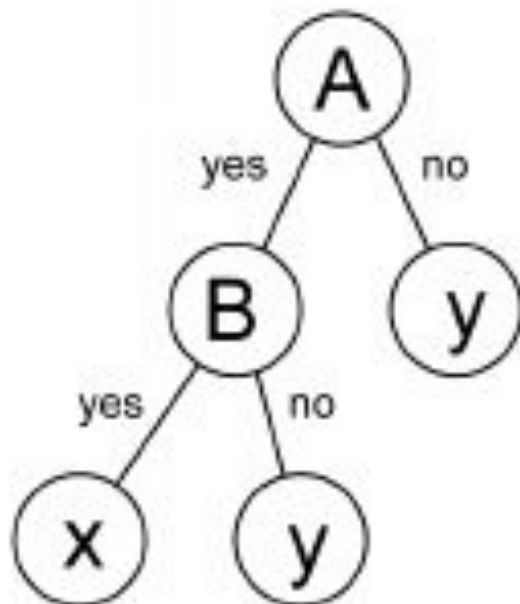
$$\begin{aligned} IG(\text{Parent}, \text{Residence}) &= E(\text{Parent}) - E(\text{Residence}) \\ &= 0.99 - 0.86 \\ &= 0.13 \end{aligned}$$

Decision Tree with Numerical Input



Rule Extraction from Decision Tree

Decision Tree:



Rule Set:

IF (A = yes) AND (B = yes)
THEN (class X)

IF (A = yes) AND (B = no)
THEN (class Y)

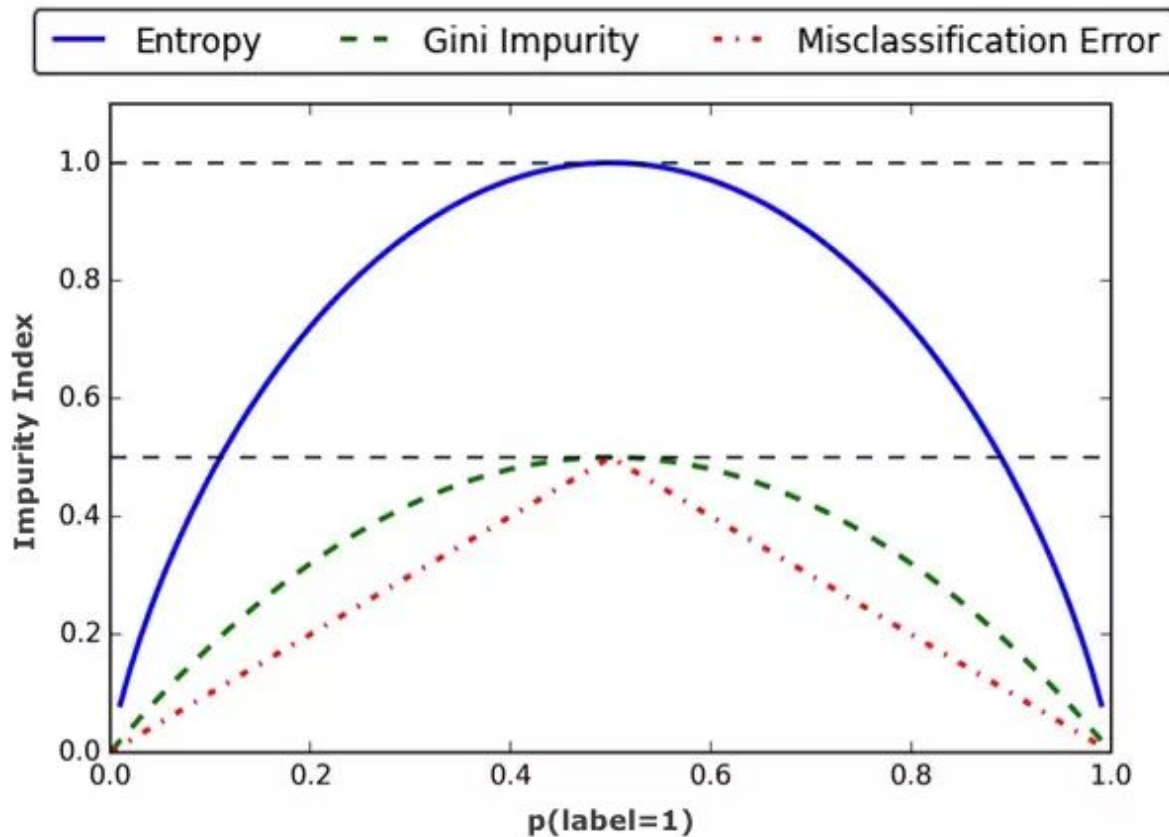
IF (A = no)
THEN (class Y)

Other Metrics for Decision Tree Classification

$$J_{Entropy} = - \sum_{i=1}^N p_i \log_2(p_i)$$

$$J_{Gini} = 1 - \sum_{i=1}^N p_i^2$$

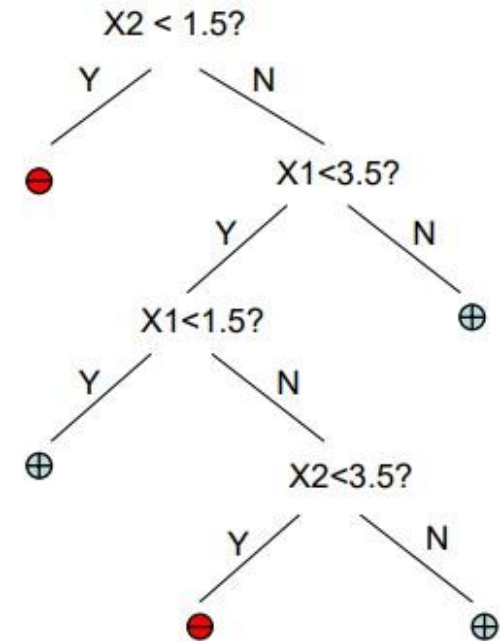
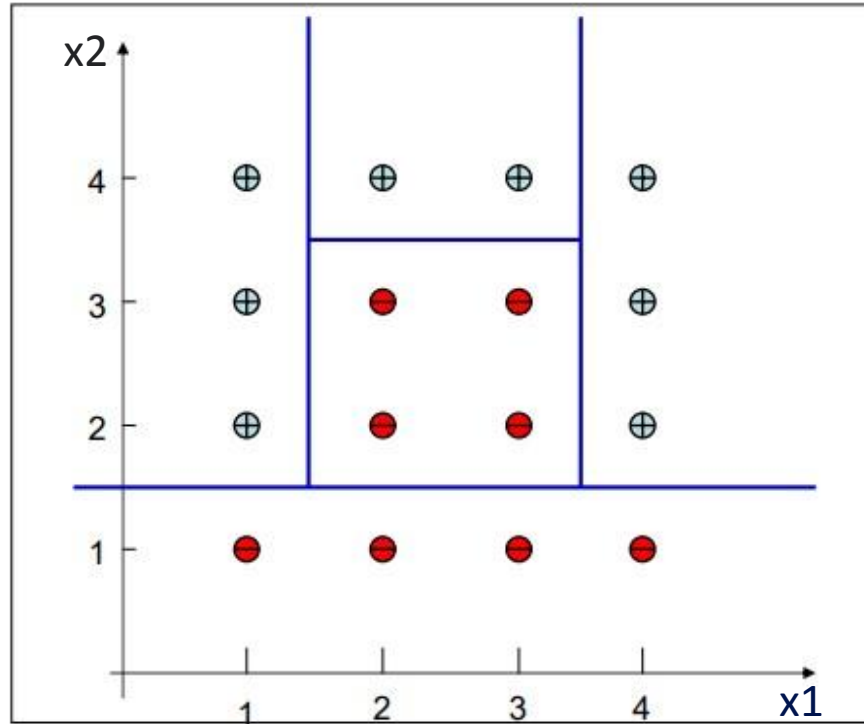
$$J_{Misclass} = 1 - \max(p_i)$$



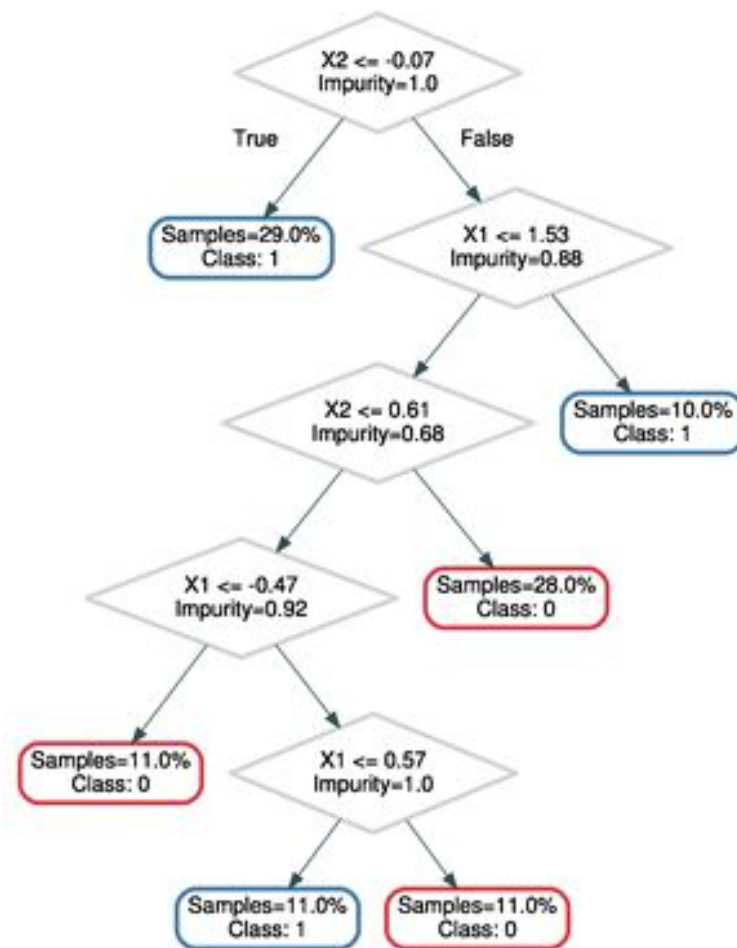
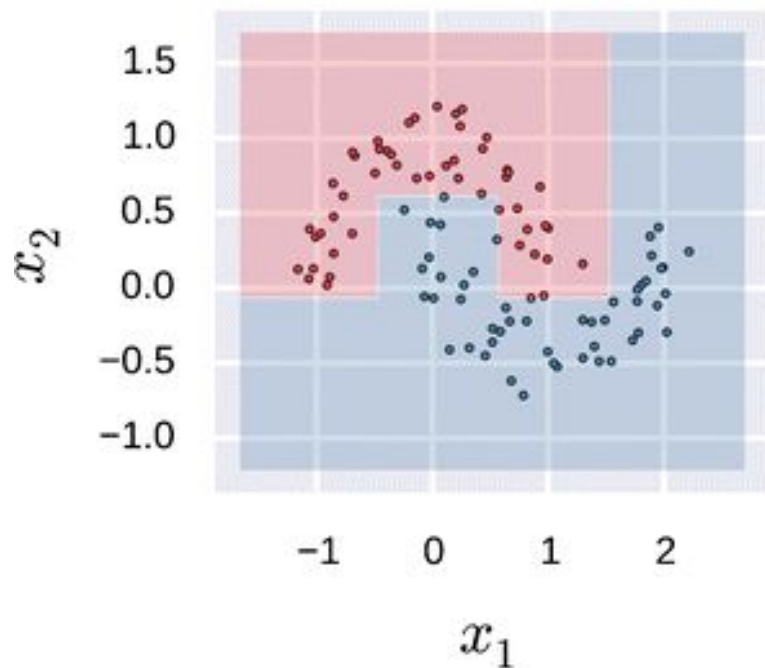
Plot for binary classification

Decision Tree and Decision Boundaries

- Decision Trees divide the input space into axis-parallel rectangles and label each rectangle with one of the K classes



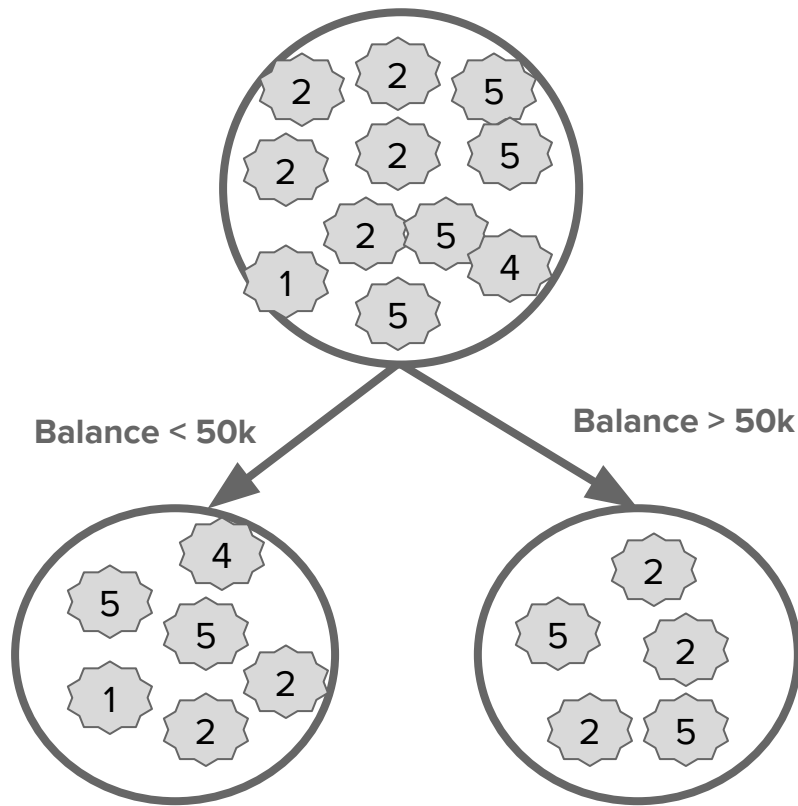
Decision Tree and Decision Boundaries



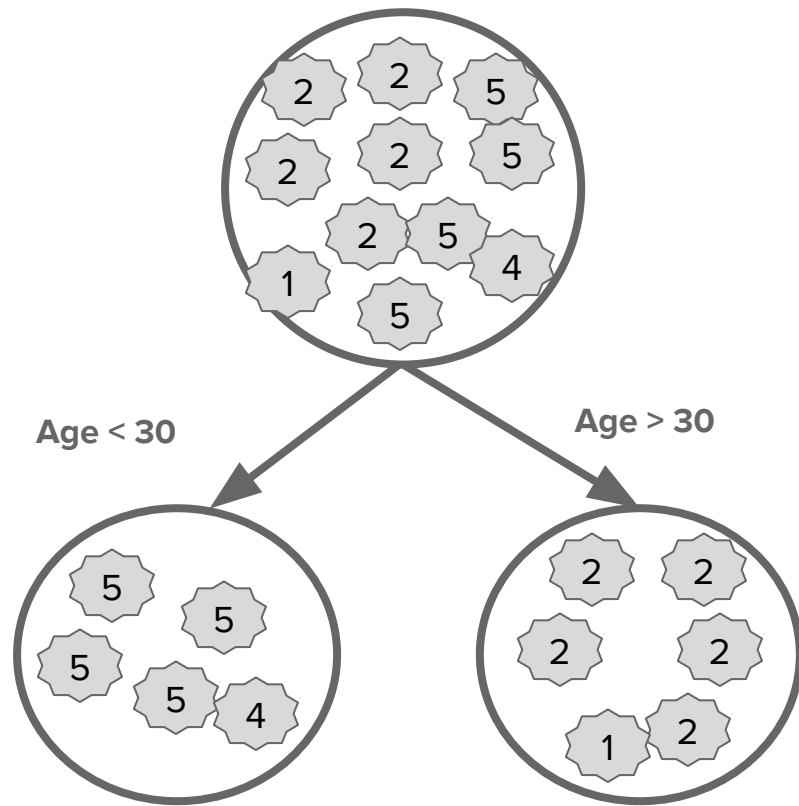
Regression Tree

Regression Tree Modeling

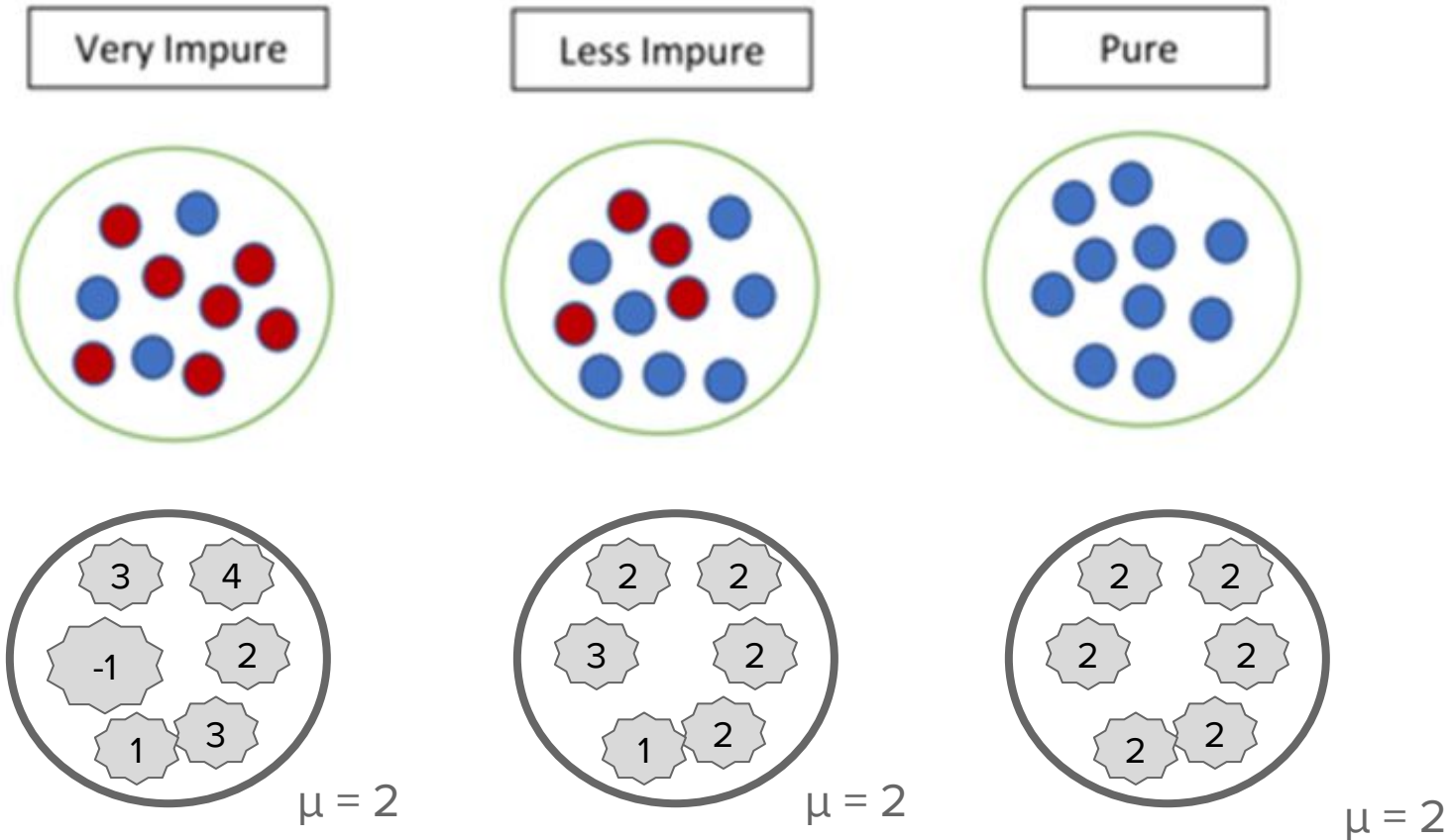
Entire population (11 instants)



Entire population (11 instants)



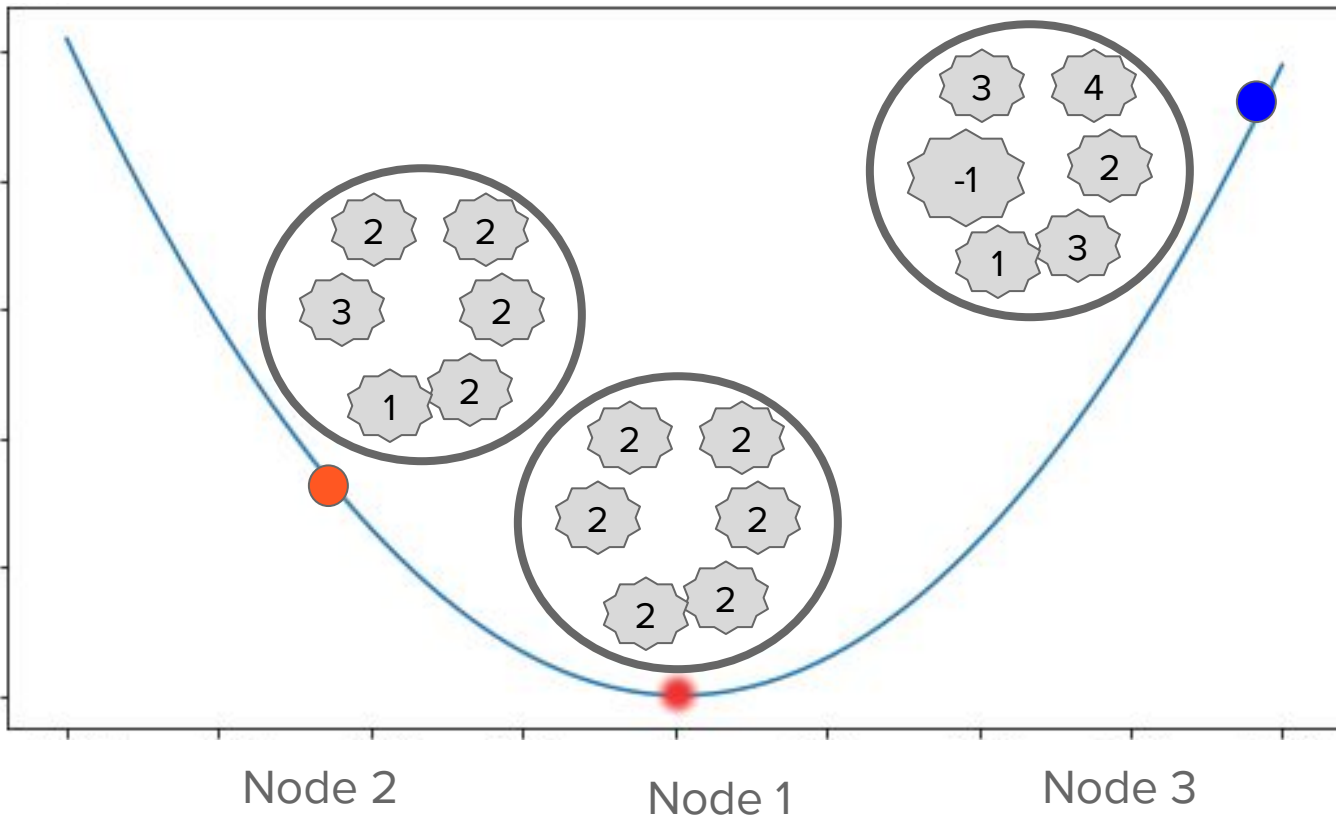
Impurities: When to split a Node?



Impurity Measure: MSE

$$MSE = \frac{1}{n} \sum \left(y - \hat{y} \right)^2$$

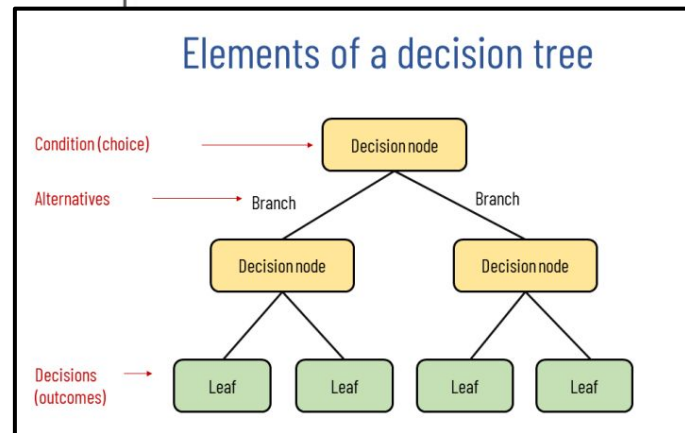
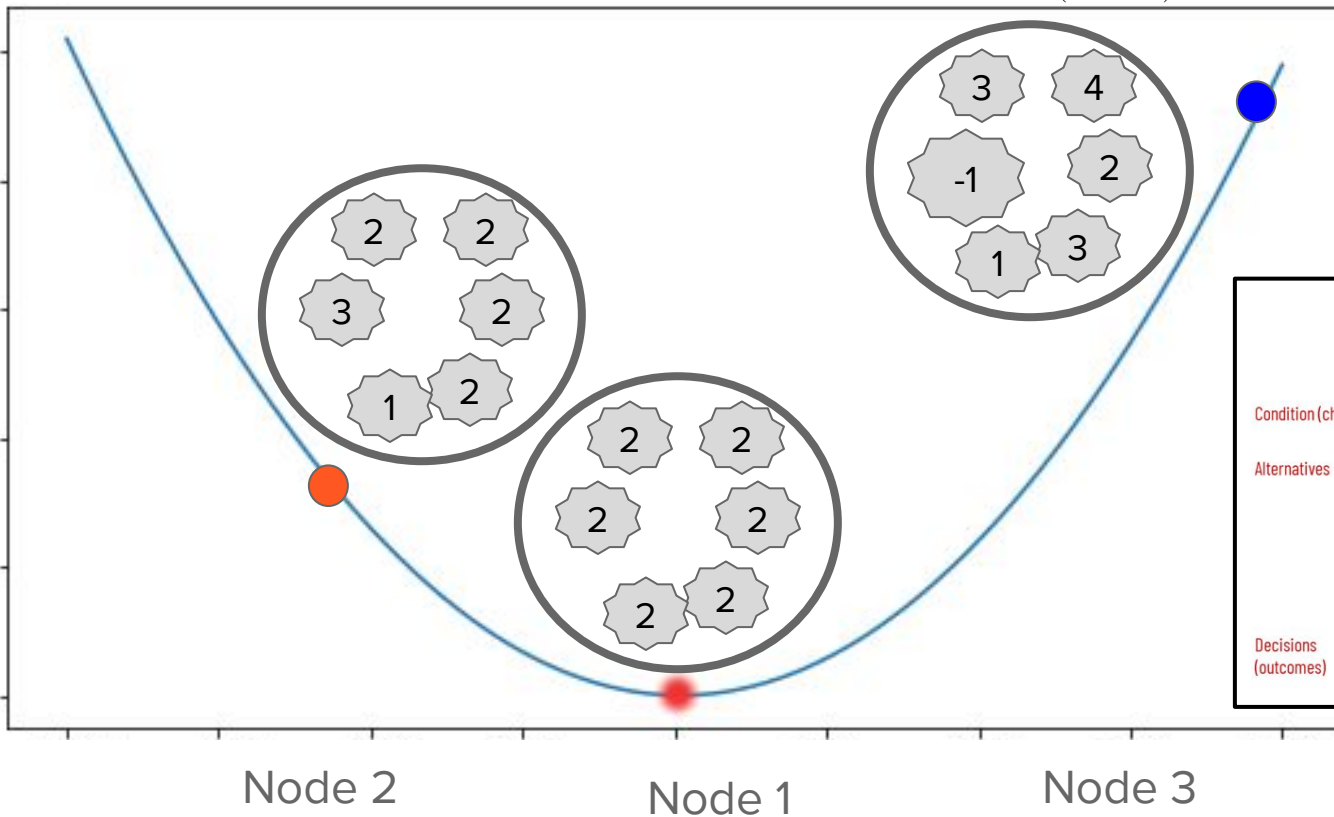
$$Gain(S, A) = MSE(S) - \sum_{v \in D_A} \frac{|S_v|}{|S|} MSE(S_v)$$



Impurity Measure: MSE

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

$$Gain(S, A) = MSE(S) - \sum_{v \in D_A} \frac{|S_v|}{|S|} MSE(S_v)$$

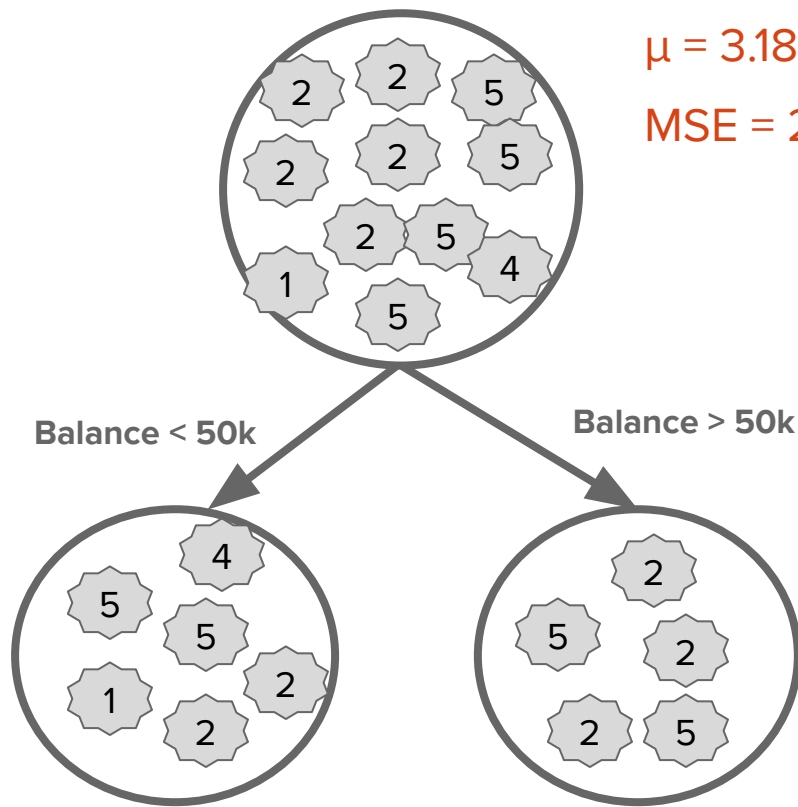


Regression Tree Modeling

Entire population (11 instants)

$$\mu = 3.182$$

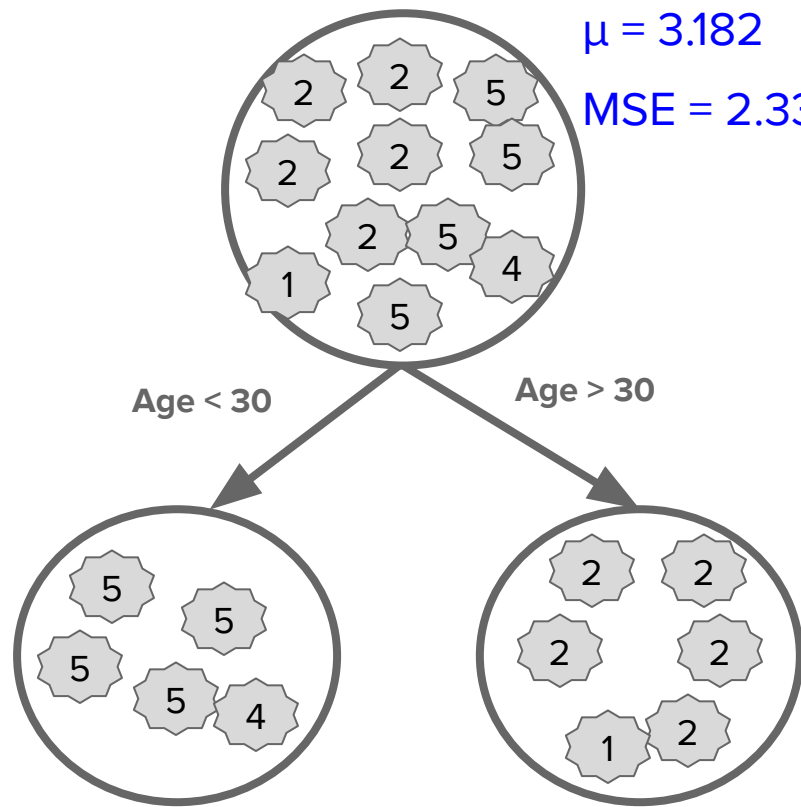
$$\text{MSE} = 2.33$$



Entire population (11 instants)

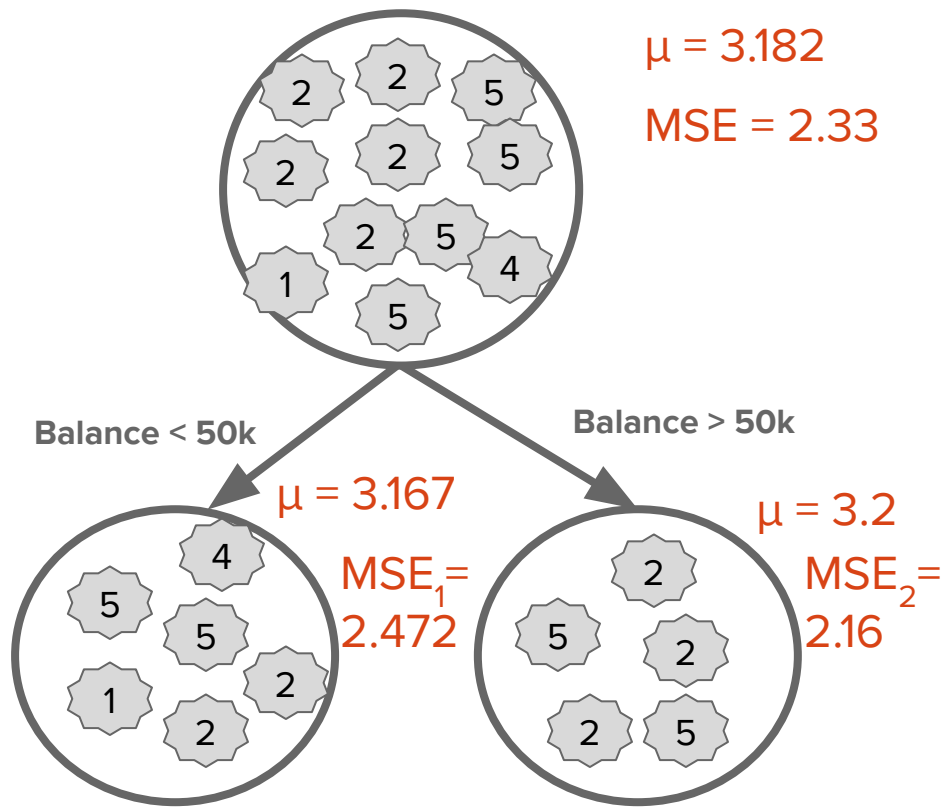
$$\mu = 3.182$$

$$\text{MSE} = 2.33$$

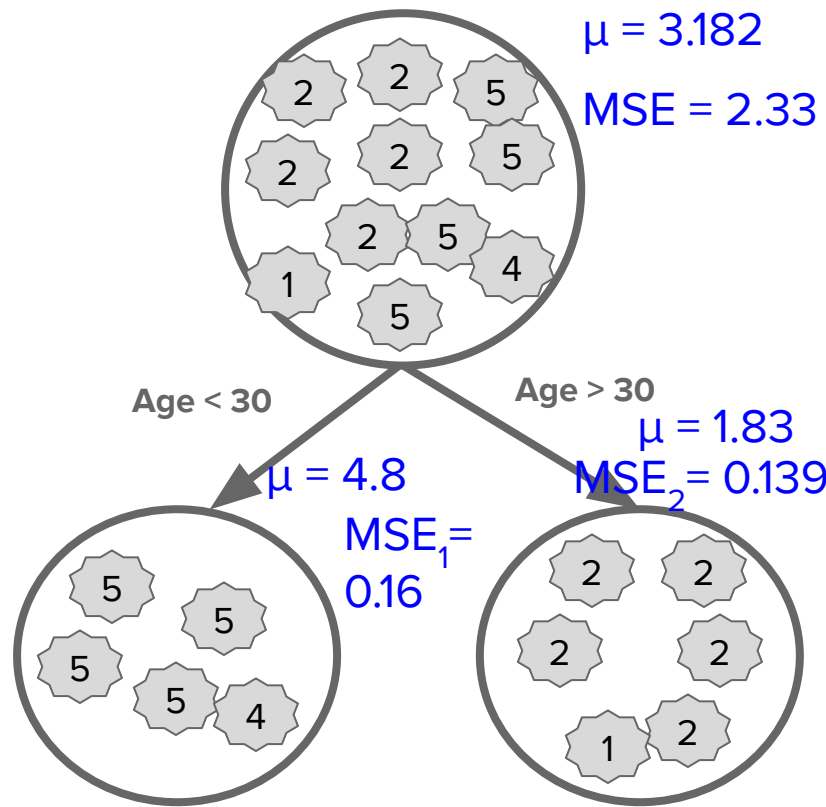


Regression Tree Modeling

Entire population (11 instants)

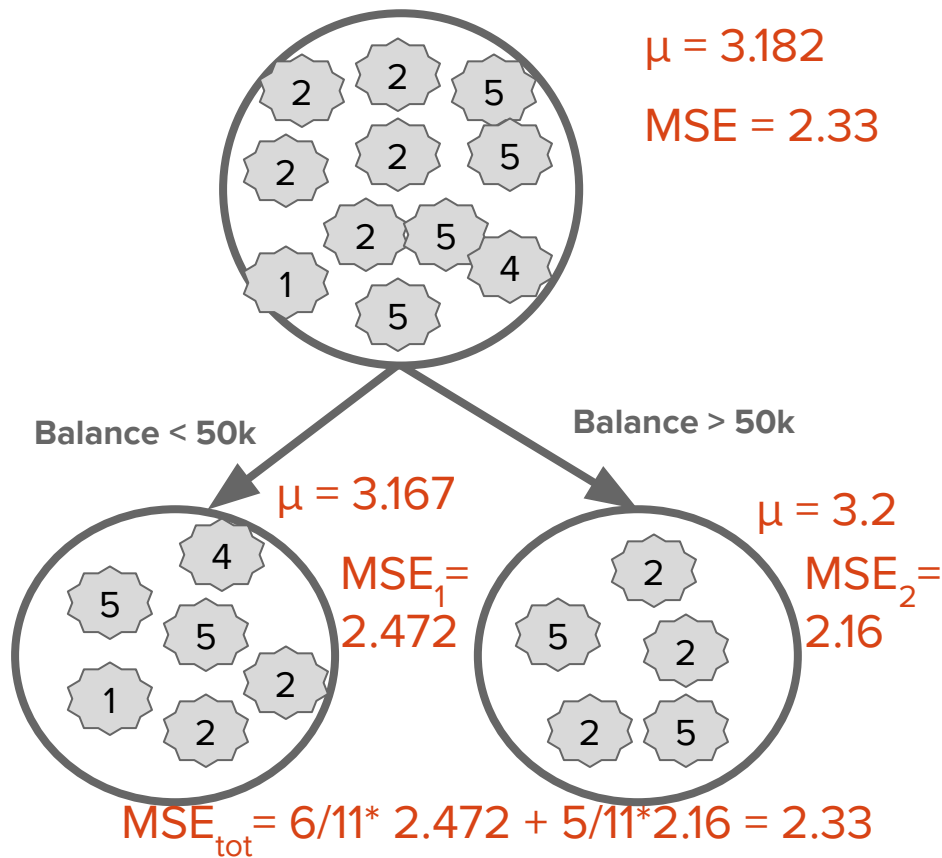


Entire population (11 instants)

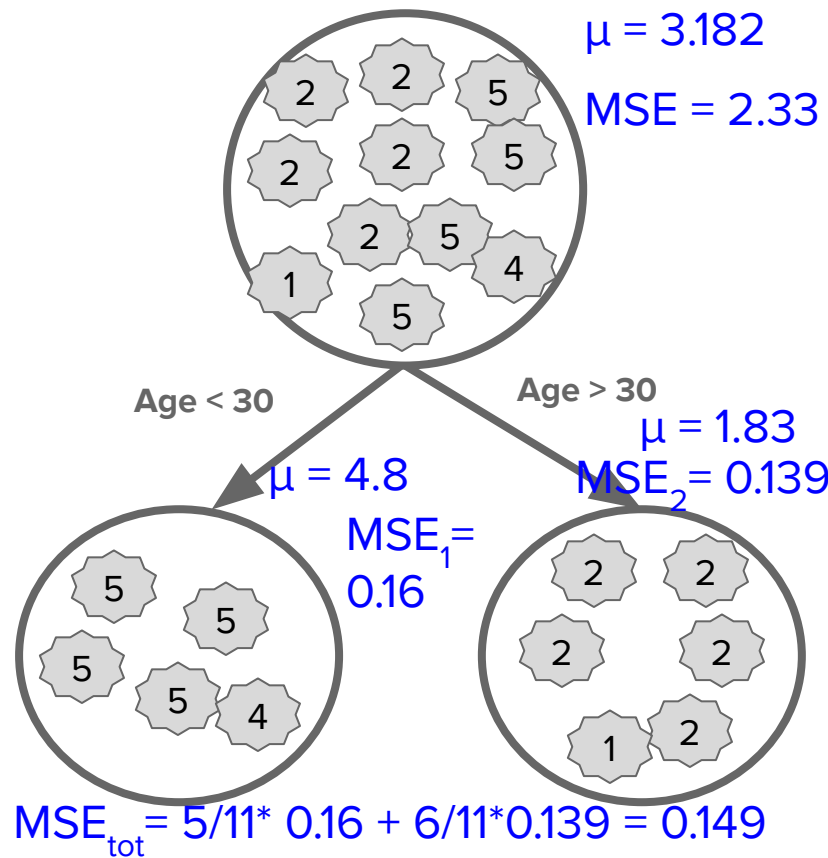


Regression Tree Modeling

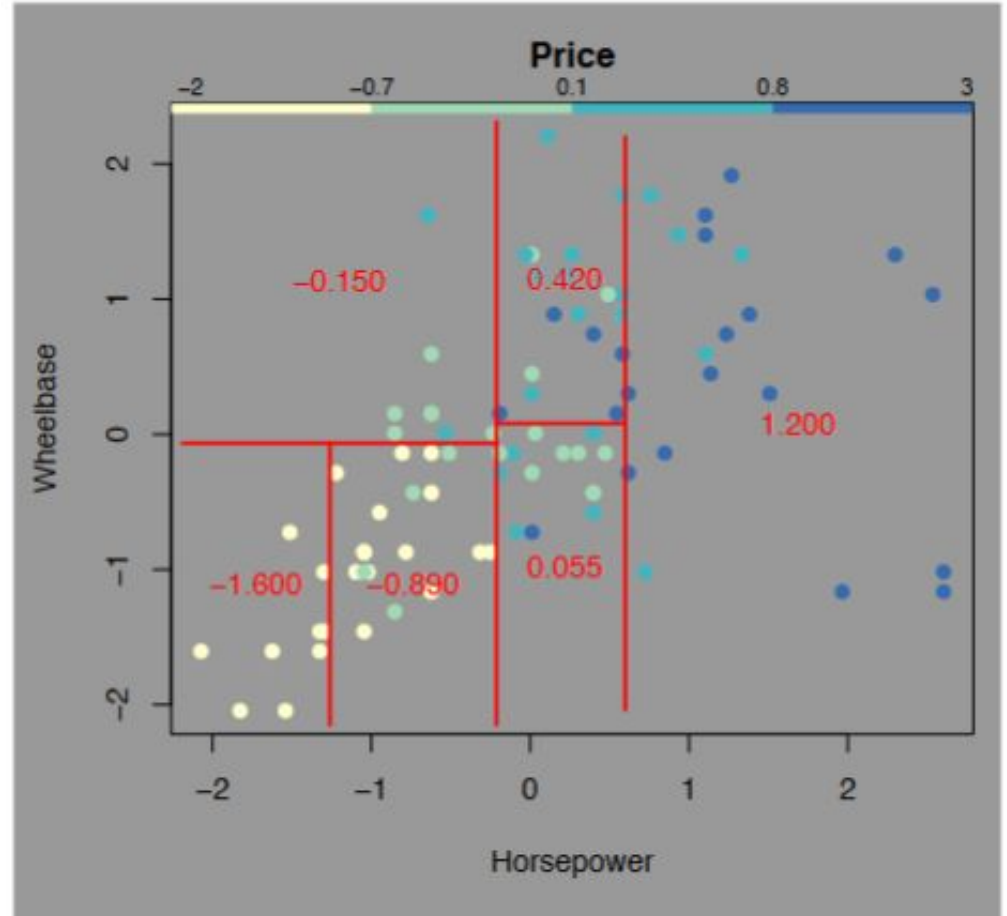
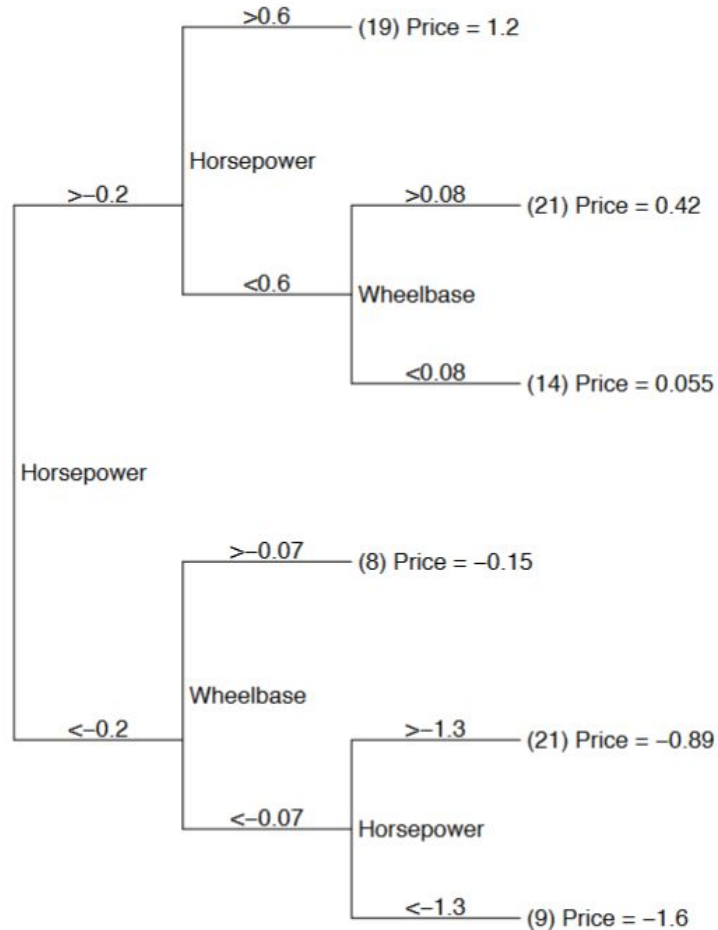
Entire population (11 instants)



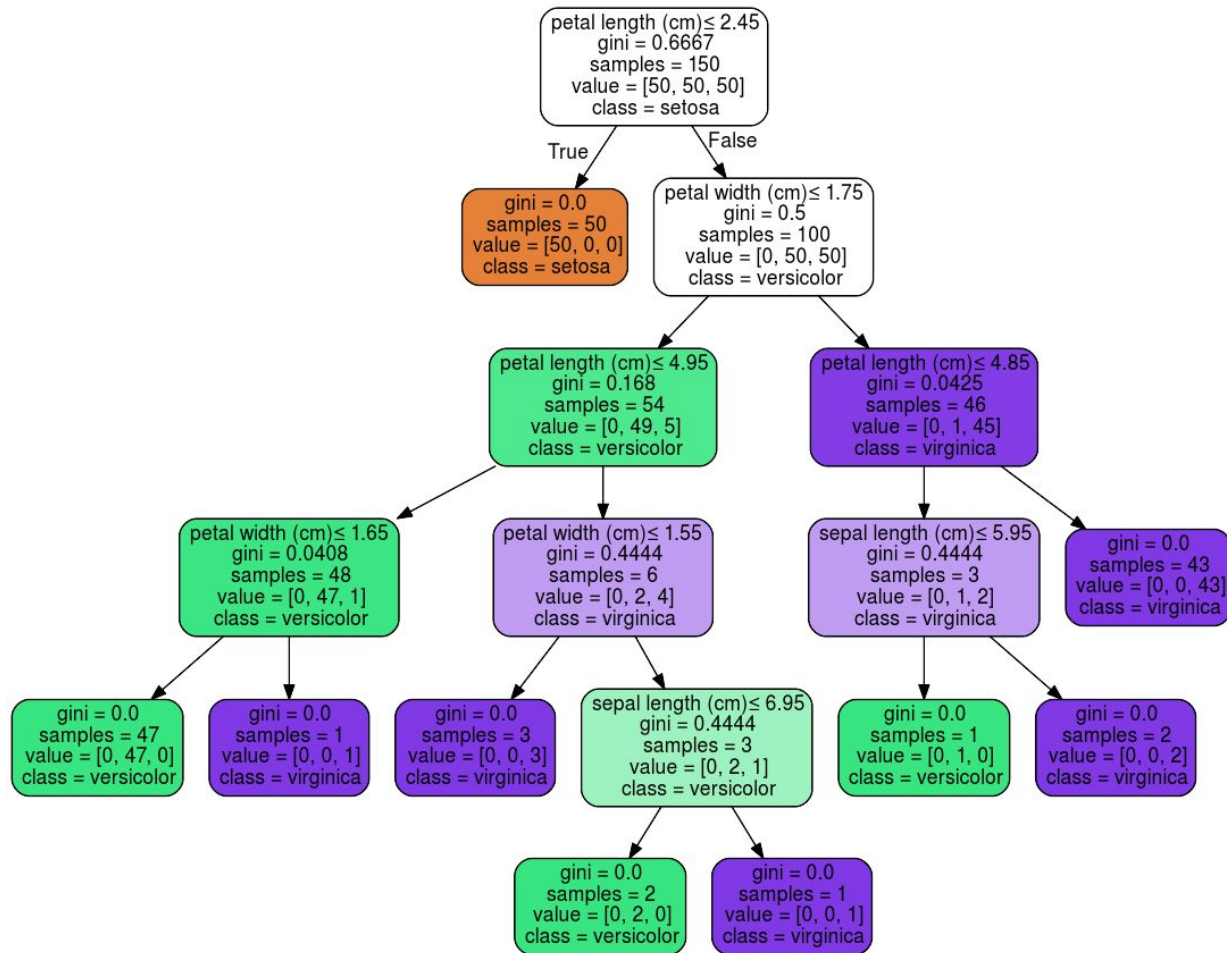
Entire population (11 instants)



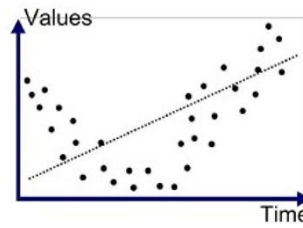
Regression Tree and Regression Boundaries



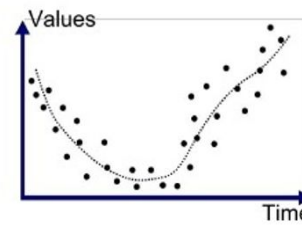
Pruning Decision Tree



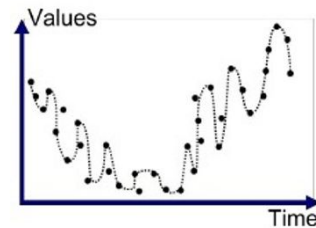
Pruning Decision Tree



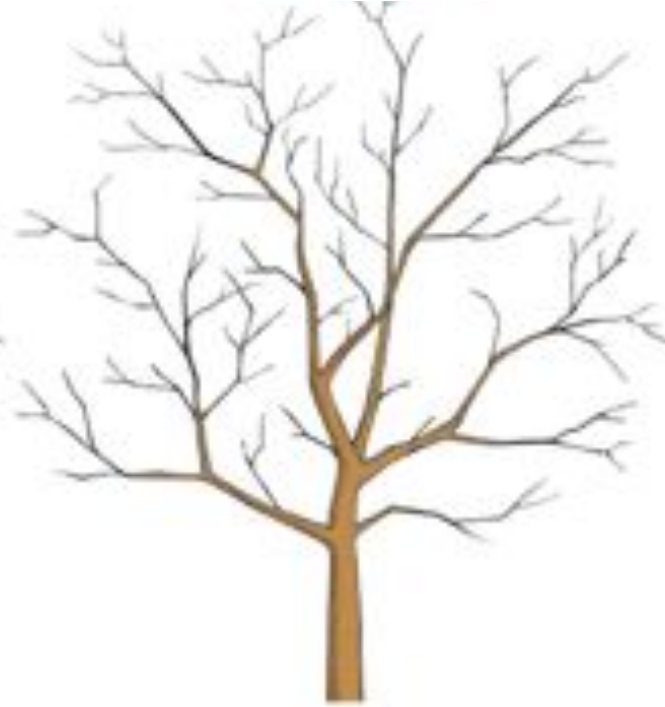
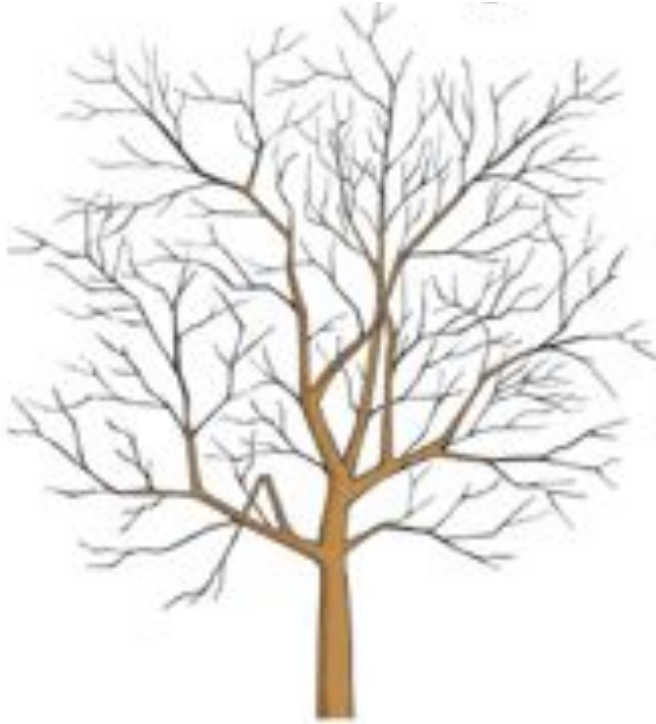
Underfitted



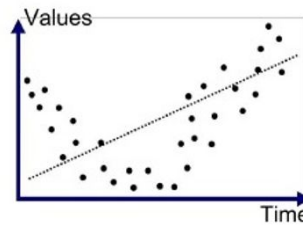
Good Fit/Robust



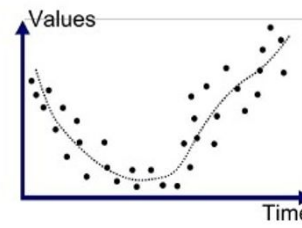
Overfitted



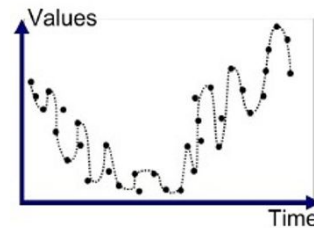
Pruning Decision Tree



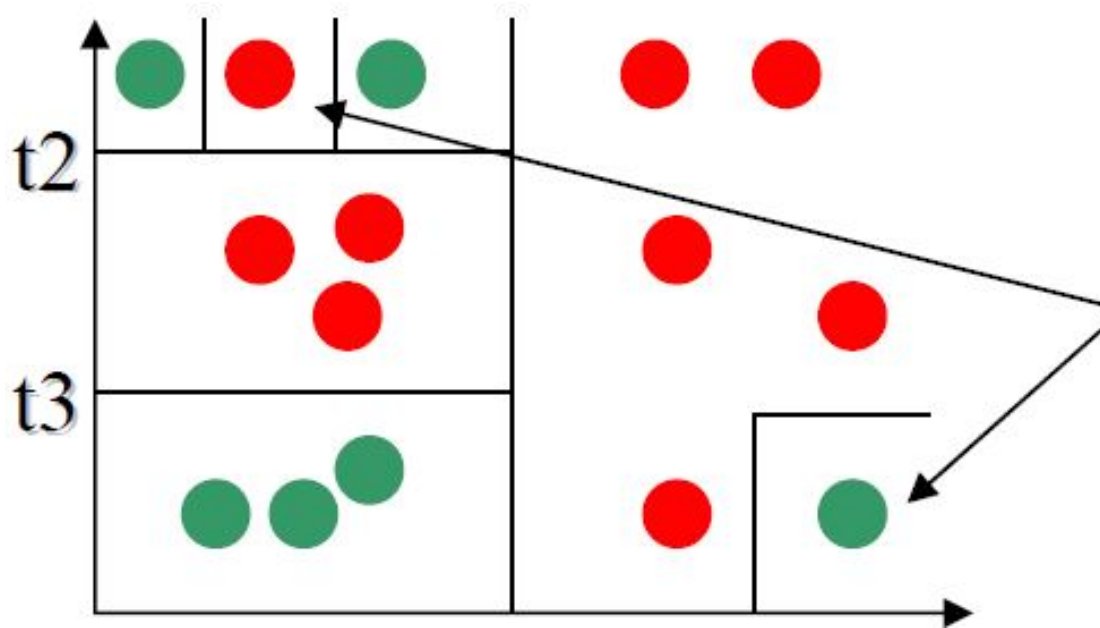
Underfitted



Good Fit/Robust

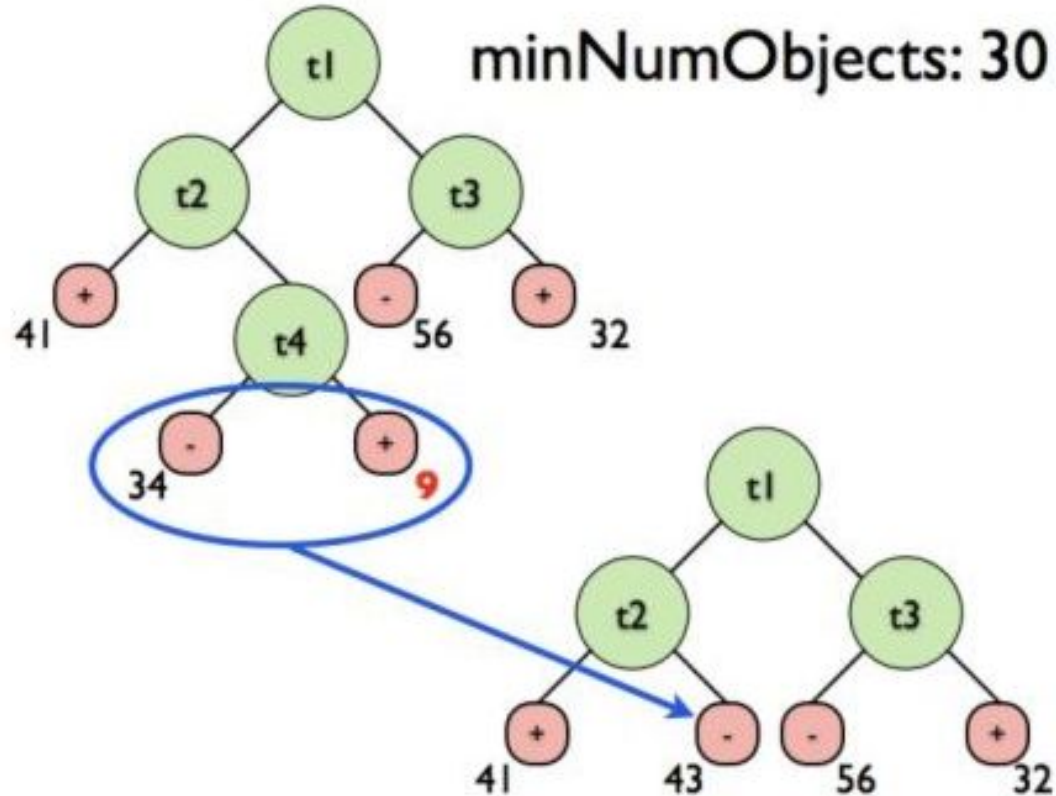


Overfitted



Possibly just noise, but the tree is grown larger to capture these examples

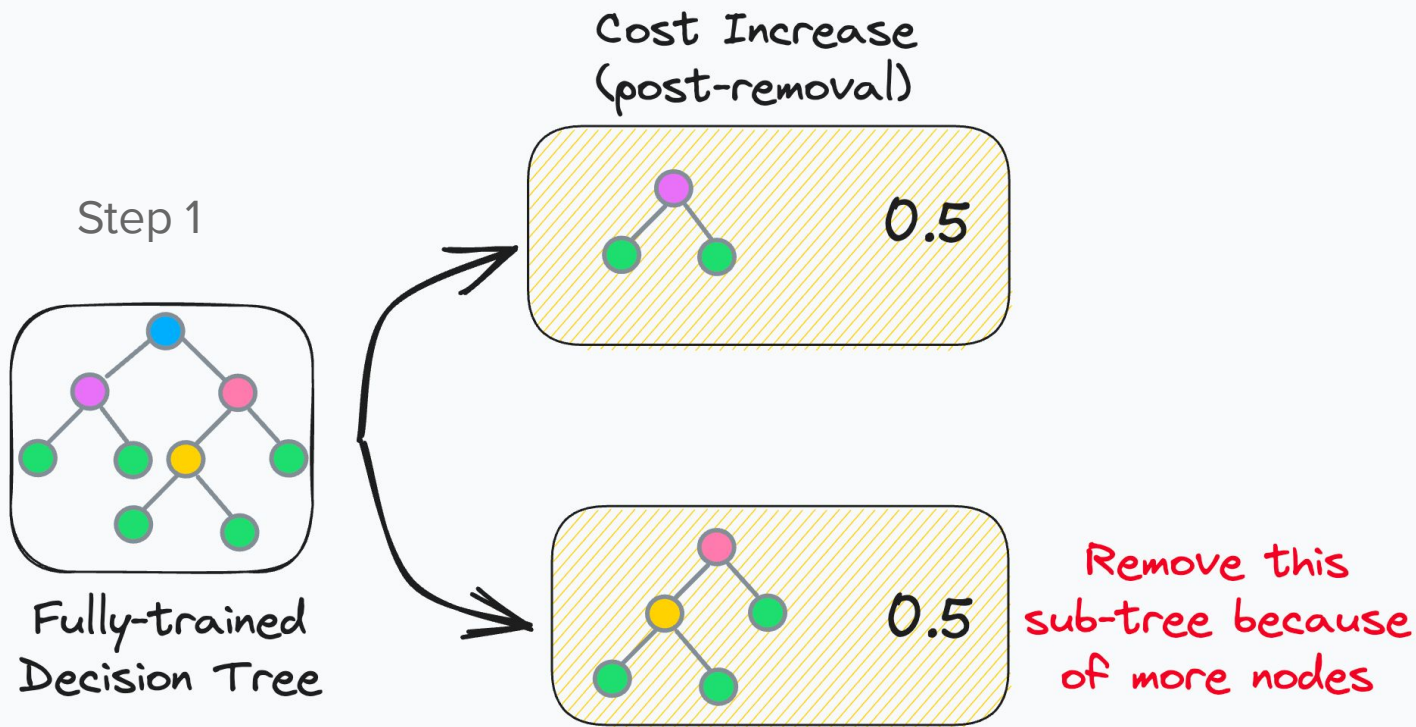
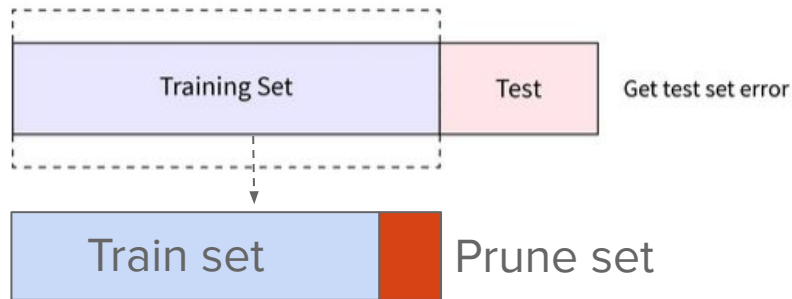
Prepruning



1. Max_depth
2. Max_leaf_nodes
3. Non-zero Entropy
4. Min Samples split
5. Min Samples Leaf

.....

Postpruning



Postpruning

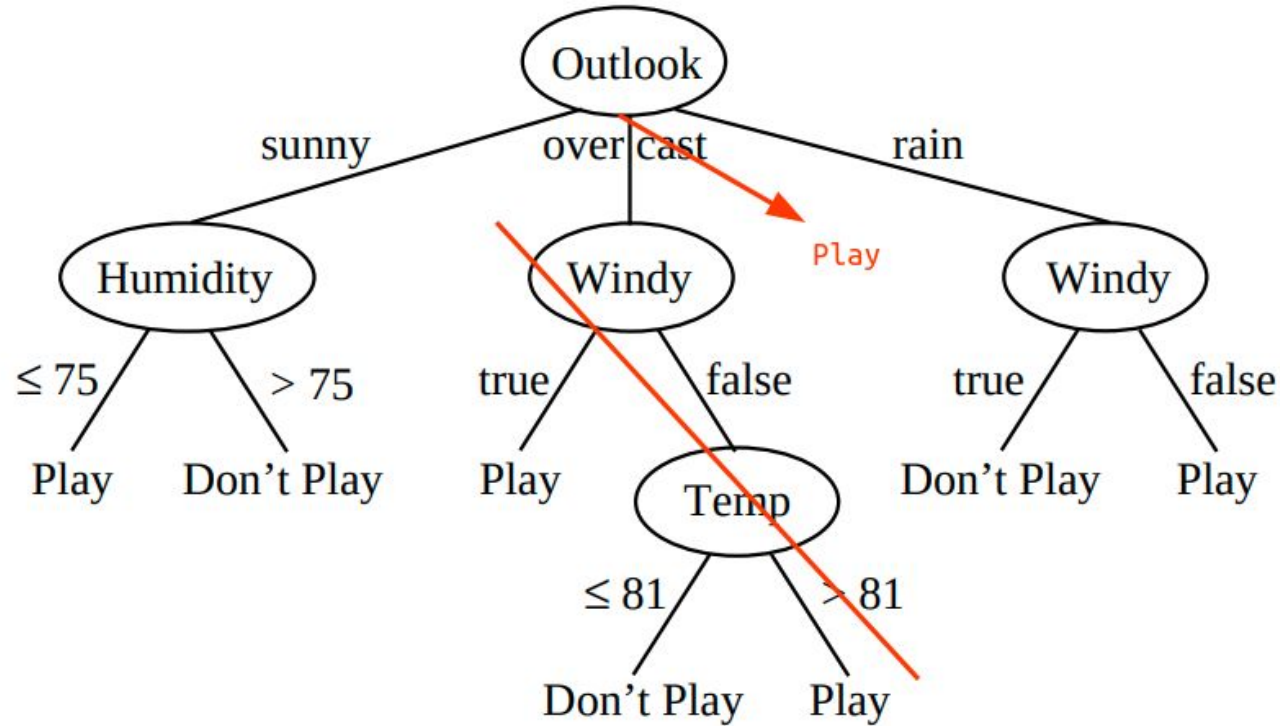


Figure 1: An example of a decision tree.